

Temporal models, novelty & section boundaries

Music as an ordered process: autocorrelation, hidden states and statistical change.

Time-series dependence

Music is nonstationary and ordered. For centered processes, lagged covariance is $C_{ab}(\ell) = E[(a_t - \mu_a)(b_{t+\ell} - \mu_b)]$. Autocorrelation identifies persistence or periodicity; cross-correlation identifies lagged association. Estimate locally when distributions change. A leading feature does not prove it causes the following event.

Autoregressive models provide simple baselines: $y_t = \sum_{j=1}^p a_j y_{t-j} + \epsilon_t$. Inspect residuals, lag selection and stability. State-space models introduce a latent trajectory: $z_{t+1} = Az_t + \eta_t$, $d_t = Cz_t + \epsilon_t$. Kalman filtering is a later option under suitable linear/Gaussian assumptions; it is not required merely because the data is a time series.

Markov and hidden Markov models

An HMM represents discrete latent musical states z_t : $p(z_{1:T}, d_{1:T}) = p(z_1) \prod_{t=2}^T p(z_t | z_{t-1}) \prod_{t=1}^T p(d_t | z_t)$. Transition probabilities encode persistence and recurrence; emissions describe feature distributions. Forward-backward estimates state marginals; Viterbi estimates a most likely path. Hidden semi-Markov models can represent section durations more explicitly.

Learned states do not automatically equal verse, chorus or bridge. Interpret them with annotations and listening.

Novelty and change-point detection

A simple boundary cue compares neighboring interval summaries, $\nu_t = \|\bar{d}_{t+} - \bar{d}_{t-}\|$. A checkerboard kernel over a self-similarity matrix can emphasize transitions from internally similar regions to dissimilar neighbors.

A statistical formulation asks whether distributions within adjacent regimes differ. It does not assume all samples before a boundary or after it share one global distribution.

A segmentation objective is $\min_{\tau_1, \dots, \tau_K} \sum_{j=0}^K \mathcal{C}(d_{\tau_j: \tau_{j+1}}) + \lambda K$. Here $\mathcal{C}$ is a within-segment cost, λ penalizes extra boundaries, and endpoints cover the recording. Minimum duration and smoothing prevent every transient from becoming a section.

Extensions include CUSUM for sequential mean shifts, likelihood ratios, kernel costs and Bayesian online change-point detection. Compare assumptions and computational costs rather than enabling every method at once.

Evaluation and interaction

Overlay candidate boundaries on playback and let the user adjust or annotate them. Compare detections against reference boundaries with declared timing tolerances and precision/recall, and inspect coarse versus fine segmentation. Hold out songs when tuning thresholds. Show uncertainty or sensitivity to settings; a novelty peak is evidence for a change, not a guaranteed musical section.

Statistical structure · Inference and experiments